



**Laboratorium
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*Institut Teknologi Sepuluh Nopember***

Laporan Akhir Praktikum Jaringan Komputer

Vlan, Trunk, OSPF MultiVendor

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1 Langkah-Langkah Percobaan

Pada modul 5 praktikum, terdapat 5 percobaan yang harus dilakukan yaitu: Konfigurasi Cisco Switch, Konfigurasi Cisco Router, Konfigurasi Huawei Router dan IP Address Antarrouter, Konfigurasi OSPF Multivendor dan OSPF Cost, serta Pengujian Failover OSPF. Percobaan dimulai dengan berikut sebagai langkah-langkahnya:

1.1 Praktikum 1 Konfigurasi Cisco Switch

1. Masuk ke terminal Cisco Switch `enable`
`conf t`
2. Membuat VLAN dengan membuat empat VLAN yaitu VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing), lalu verifikasi dengan perintah `show vlan brief`.
3. Pastikan VLAN sudah dibuat dan statusnya aktif dengan mengecek output dari `show vlan brief`.
4. Konfigurasi Access port untuk setiap VLAN sesuai dengan interface pada Cisco Switch.
 - Interface `gi0/2` dikonfigurasi sebagai access port untuk VLAN 10 (Finance).
 - Interface `gi0/1` dikonfigurasi sebagai access port untuk VLAN 20 (HR).
 - Interface `gi1/1` dikonfigurasi sebagai access port untuk VLAN 30 (IT).
 - Interface `gi0/3` dikonfigurasi sebagai access port untuk VLAN 40 (Marketing).
5. Konfigurasi Trunk ke Cisco Router dengan mengatur interface `gi0/0` sebagai trunk port yang membawa VLAN 10, 20, 30, dan 40 menuju Cisco Router menggunakan enkapsulasi `dot1q`. Verifikasi dengan perintah `show interfaces trunk`.
6. Simpan semua konfigurasi dengan perintah `write memory`.

1.2 Praktikum 2 Konfigurasi Cisco Router

1. Masuk ke terminal Cisco Switch `enable`
`conf t`
2. Konfigurasi Interface Trunk ke Switch dengan mengaktifkan interface `gig0/2` yang terhubung ke Cisco Switch tanpa IP address dan mengaktifkannya dengan `no shutdown`.
3. Konfigurasi Subinterface VLAN 10 pada `gig0/2.10` dengan enkapsulasi `dot1Q 10` dan IP address `192.168.10.1/24` sebagai gateway VLAN 10 (Finance).
4. Konfigurasi Subinterface VLAN 20 pada `gig0/2.20` dengan enkapsulasi `dot1Q 20` dan IP address `192.168.20.1/24` sebagai gateway VLAN 20 (HR).
5. Konfigurasi Subinterface VLAN 30 pada `gig0/2.30` dengan enkapsulasi `dot1Q 30` dan IP address `192.168.30.1/24` sebagai gateway VLAN 30 (IT).
6. Konfigurasi Subinterface VLAN 40 pada `gig0/2.40` dengan enkapsulasi `dot1Q 40` dan IP address `192.168.40.1/24` sebagai gateway VLAN 40 (Marketing).

7. Konfigurasi DHCP-Server untuk VLAN 10 dengan mengecualikan IP 192.168.10.1-192.168.10.10 dan membuat pool VLAN10-FINANCE dengan network 192.168.10.0/24 dan DNS server 8.8.8.8.
8. Konfigurasi DHCP-Server untuk VLAN 20 dengan mengecualikan IP 192.168.20.1-192.168.20.10 dan membuat pool VLAN20-HR dengan network 192.168.20.0/24 dan DNS server 8.8.8.8.
9. Konfigurasi IP Static pada VPCS untuk VLAN 30 dan VLAN 40.
 - PC VLAN 30 diatur dengan IP 192.168.30.10/24 dan gateway 192.168.30.1.
 - PC VLAN 40 diatur dengan IP 192.168.40.10/24 dan gateway 192.168.40.1.
 - PC VLAN 10 dan VLAN 20 menggunakan DHCP dengan perintah `ip dhcp`.
10. Verifikasi Cisco Router dengan mengecek interface menggunakan `show ip interface brief`, DHCP pool dengan `show ip dhcp pool`, dan binding DHCP dengan `show ip dhcp binding`.
11. Pengujian dari VPCS dengan melakukan request IP DHCP pada PC VLAN 10 dan VLAN 20, lalu ping ke masing-masing gateway, dan uji ping antar-VLAN dari VLAN 10 ke VLAN 20, 30, dan 40.

1.3 Praktikum 3 Konfigurasi Huawei Router dan IP Address Antarrouter

1. Konfigurasi LAN Huawei dengan mengatur IP address 192.168.50.1/24 pada interface Ethernet1/0/2 sebagai gateway LAN Huawei, lalu diakhiri dengan perintah `commit`.
2. Konfigurasi IP VPC LAN Huawei dengan mengatur IP statis 192.168.50.10/24 dengan gateway 192.168.50.1 pada PC LAN Huawei, lalu verifikasi dengan `show ip` dan uji ping ke gateway 192.168.50.1.
3. Konfigurasi IP link Cisco ↔ Huawei Link 1 menggunakan network 10.10.10.0/30.
 - Cisco Router: interface `gi0/3` dikonfigurasi dengan IP 10.10.10.1/30.
 - Huawei Router: interface `Ethernet1/0/0` dikonfigurasi dengan IP 10.10.10.2/30.
4. Konfigurasi IP Link Cisco ↔ Huawei Link 2 menggunakan network 10.10.11.0/30.
 - Cisco Router: interface `gi0/0` dikonfigurasi dengan IP 10.10.11.1/30.
 - Huawei Router: interface `Ethernet1/0/4` dikonfigurasi dengan IP 10.10.11.2/30.
5. Konfigurasi IP link Huawei ↔ MikroTik menggunakan network 10.10.20.0/30.
 - Huawei Router: interface `Ethernet1/0/3` dikonfigurasi dengan IP 10.10.20.1/30.
 - MikroTik: interface `ether1` dikonfigurasi dengan IP 10.10.20.2/30.
6. Konfigurasi IP link Cisco ↔ MikroTik menggunakan network 10.10.30.0/30.
 - Cisco Router: interface `gi0/1` dikonfigurasi dengan IP 10.10.30.1/30.
 - MikroTik: interface `ether2` dikonfigurasi dengan IP 10.10.30.2/30.
7. Verifikasi Interface pada masing-masing perangkat.
 - Cisco Router: `show ip interface brief`.

- Huawei Router: `display ip interface brief`.
 - MikroTik: `/ip address print`.
8. Verifikasi ping Antar Router untuk memastikan semua link antarrouter sudah aktif dan dapat saling berkomunikasi.
- Cisco Router ping ke 10.10.10.2, 10.10.11.2, dan 10.10.30.2.
 - Huawei Router ping ke 10.10.10.1, 10.10.11.1, dan 10.10.20.2.
 - MikroTik ping ke 10.10.20.1 dan 10.10.30.1.

1.4 Praktikum 4 Konfigurasi OSPF Multivendor dan OSPF Cost

1. Konsep routing OSPF pada topologi ini menempatkan dua link Cisco–Huawei sebagai jalur utama dengan cost 10, sedangkan MikroTik berfungsi sebagai jalur cadangan dengan cost 100.
2. Konfigurasi OSPF pada Cisco Router dengan router-id 1.1.1.1, mengiklankan semua network VLAN dan link antarrouter ke area 0, serta mengatur OSPF cost interface menuju Huawei menjadi 10 dan menuju MikroTik menjadi 100.
3. Konfigurasi OSPF pada Huawei Router dengan router-id 2.2.2.2, mengiklankan semua network LAN dan link antarrouter ke area 0.0.0.0, serta mengatur OSPF cost interface menuju Cisco menjadi 10 dan menuju MikroTik menjadi 100. Verifikasi dengan `display ospf peer` dan `display ip routing-table protocol ospf`.
4. Konfigurasi OSPF pada MikroTik dengan router-id 3.3.3.3, menambahkan network 10.10.20.0/30 dan 10.10.30.0/30 ke area backbone, serta mengatur cost interface ether1 dan ether2 menjadi 100 sebagai jalur backup.
5. Verifikasi OSPF Neighbor pada ketiga perangkat.
 - Cisco Router: `show ip ospf neighbor` — Huawei muncul dua kali, MikroTik muncul satu kali.
 - Huawei Router: `display ospf peer` — Cisco muncul dua kali, MikroTik muncul satu kali.
 - MikroTik: `/routing ospf neighbor print` — Cisco dan Huawei muncul.
6. Verifikasi Routing Table pada ketiga perangkat untuk memastikan semua network telah didistribusikan melalui OSPF.
 - Cisco Router: `show ip route 192.168.50.0` dan `show ip route ospf`.
 - Huawei Router: `display ip routing-table protocol ospf`.
 - MikroTik: `/ip route print where ospf`.
7. Verifikasi Koneksi end-to-end dengan melakukan ping dari PC VLAN 10, 20, 30, dan 40 menuju PC LAN Huawei di 192.168.50.10, serta ping balik dari PC LAN Huawei ke semua network VLAN Cisco.

1.5 Praktikum 5 Pengujian Failover OSPF

1. Kondisi normal: Dua link Cisco–Huawei aktif dengan mengecek route ke 192.168.50.0 menggunakan `show ip route 192.168.50.0` dan melakukan `trace 192.168.50.10` dari PC VLAN untuk memastikan traffic melewati jalur langsung Cisco → Huawei.
2. Kondisi 1 link Cisco–Huawei mati dengan mematikan interface `gi0/3` menggunakan perintah `shutdown`, lalu cek ulang routing table dan lakukan `traceroute`. Traffic harus tetap melewati jalur langsung Cisco melalui link kedua.
3. Kondisi dua link Cisco–Huawei mati dengan mematikan juga interface `gi0/0`, lalu cek routing table dan lakukan `traceroute`. Route harus berpindah ke jalur backup melalui MikroTik dengan jalur PC VLAN → Cisco Router → MikroTik → Huawei Router → LAN Huawei.
4. Mengaktifkan kembali dua link Cisco–Huawei dengan perintah `no shutdown` pada interface `gi0/3` dan `gi0/0`, lalu verifikasi routing table dan `traceroute` untuk memastikan jalur routing kembali ke link utama Cisco–Huawei.

2 Analisis Hasil Percobaan

Pada praktikum modul 5 ini, seluruh konfigurasi mulai dari VLAN dan inter-VLAN routing pada Cisco Switch dan Router, pembangunan link antarrouter multivendor (Cisco, Huawei, MikroTik), hingga distribusi routing dinamis OSPF berhasil dilakukan sesuai teori. Mekanisme OSPF cost terbukti berfungsi dengan benar, di mana dua link langsung Cisco–Huawei dengan cost 10 selalu diprioritaskan sebagai jalur utama dibandingkan jalur melalui MikroTik yang memiliki cost 100. Pengujian failover juga membuktikan bahwa OSPF mampu melakukan konvergensi secara otomatis ketika kedua link utama dimatikan, sehingga traffic beralih ke jalur backup melalui MikroTik tanpa perlu konfigurasi manual tambahan.

3 Hasil Tugas Modul

<https://github.com/paaiz/Praktikum-Jarkom-2026-Kelompok-06/blob/5a1245c3e3861c3464660d7fff49962e7e9695/LA/README.md>

4 Kesimpulan

Dari modul 5 praktikum, praktikan berhasil memahami implementasi VLAN, inter-VLAN routing, dan OSPF multivendor secara end-to-end. Konfigurasi OSPF cost terbukti mampu menentukan jalur utama dan backup secara otomatis, serta mekanisme failover OSPF berjalan sesuai teori ketika link utama mengalami kegagalan, sehingga seluruh tujuan praktikum tercapai.

5 Lampiran

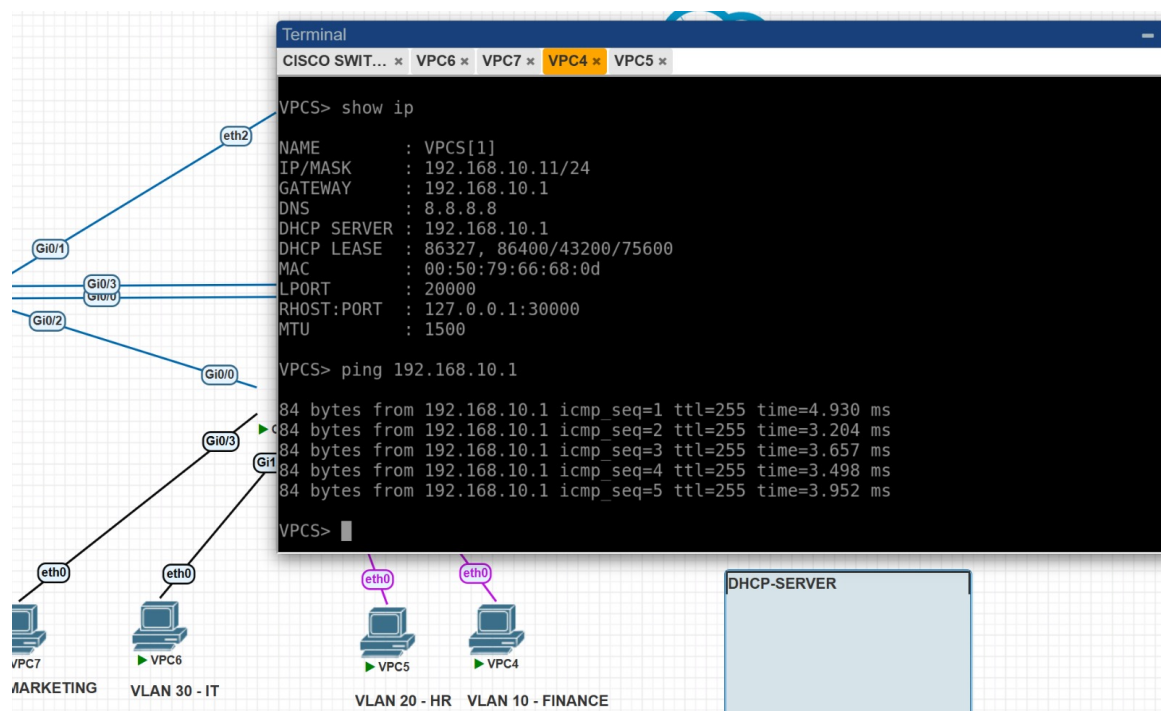
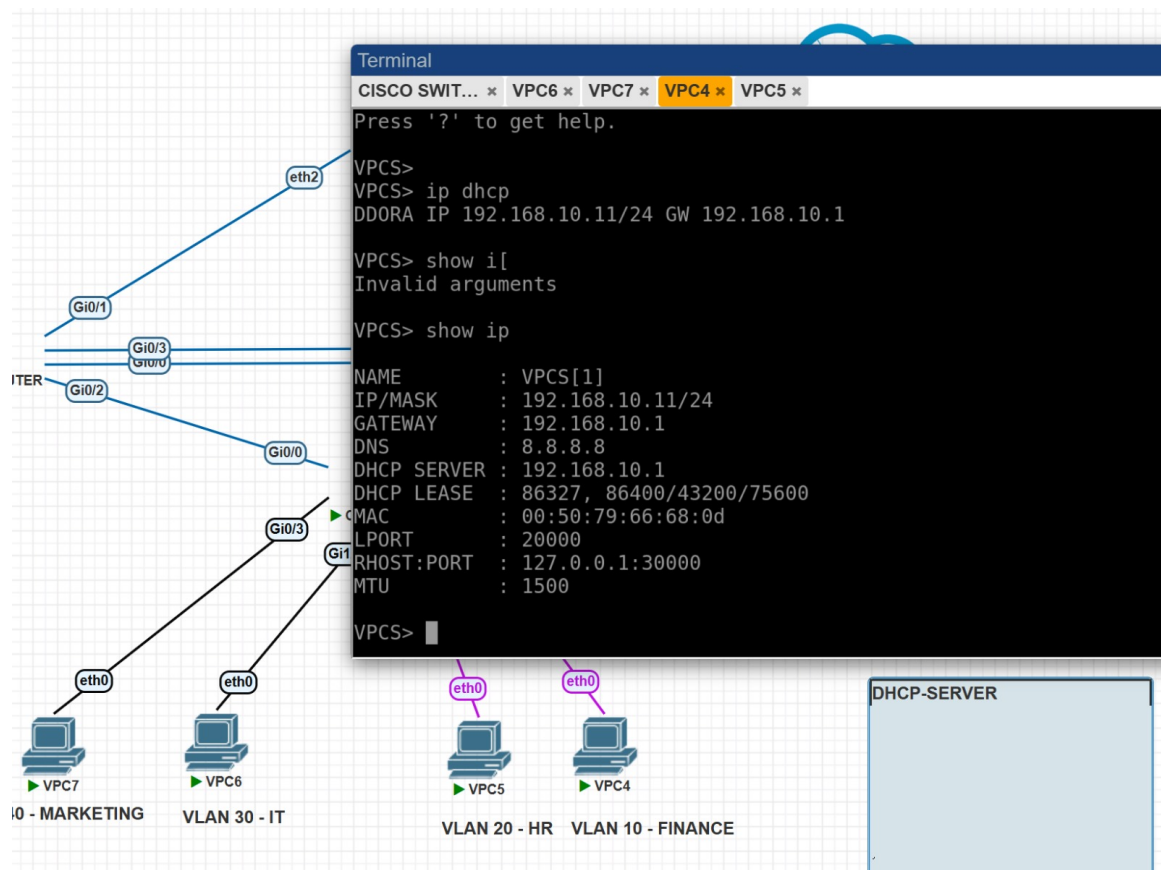
```
*Jun  6 03:42:42.862: %SYS-5-CONFIG_I: Configured from console by c
Router#
Router#show ip interface brief
Interface                IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0       unassigned      YES unset  administratively down  down
GigabitEthernet0/1       unassigned      YES unset  administratively down  down
GigabitEthernet0/2       unassigned      YES unset  up                    up
GigabitEthernet0/2.10    192.168.10.1    YES manual  up                    up
GigabitEthernet0/2.20    192.168.20.1    YES manual  up                    up
GigabitEthernet0/2.30    192.168.30.1    YES manual  up                    up
GigabitEthernet0/2.40    192.168.40.1    YES manual  up                    up
GigabitEthernet0/3       unassigned      YES unset  administratively down  down
Router#
Router#
```

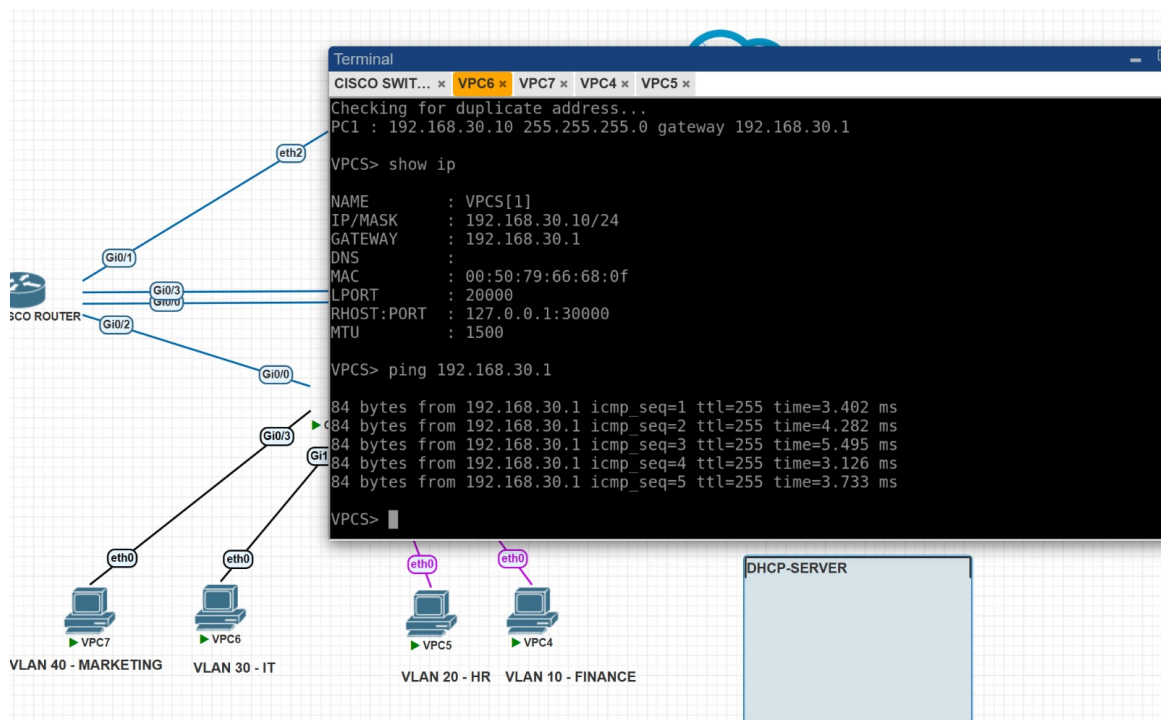
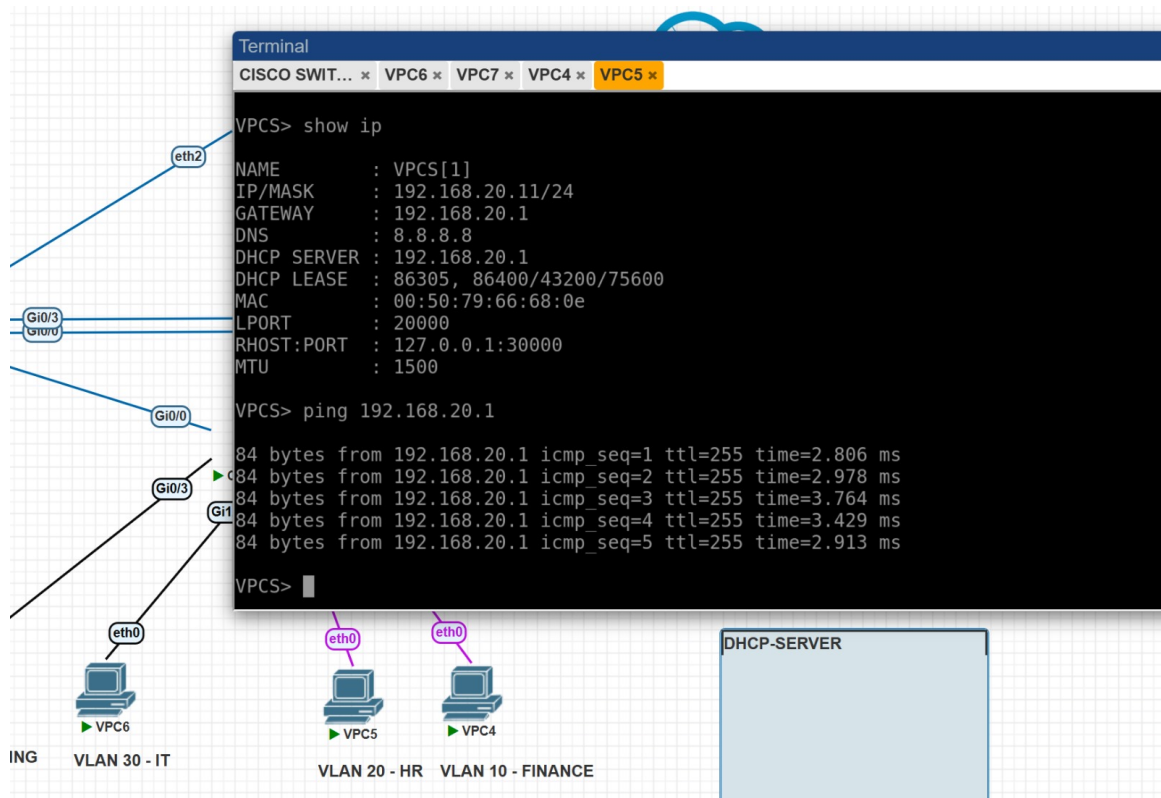
```
Router#
Router#show ip dhcp pool

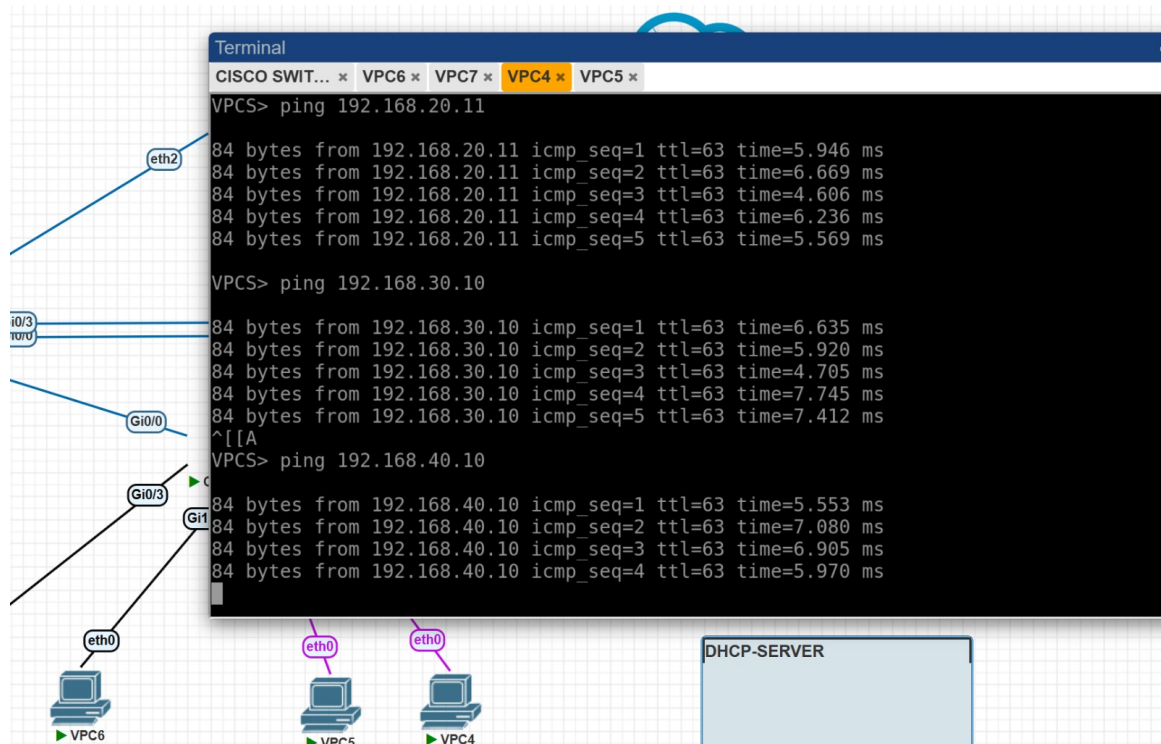
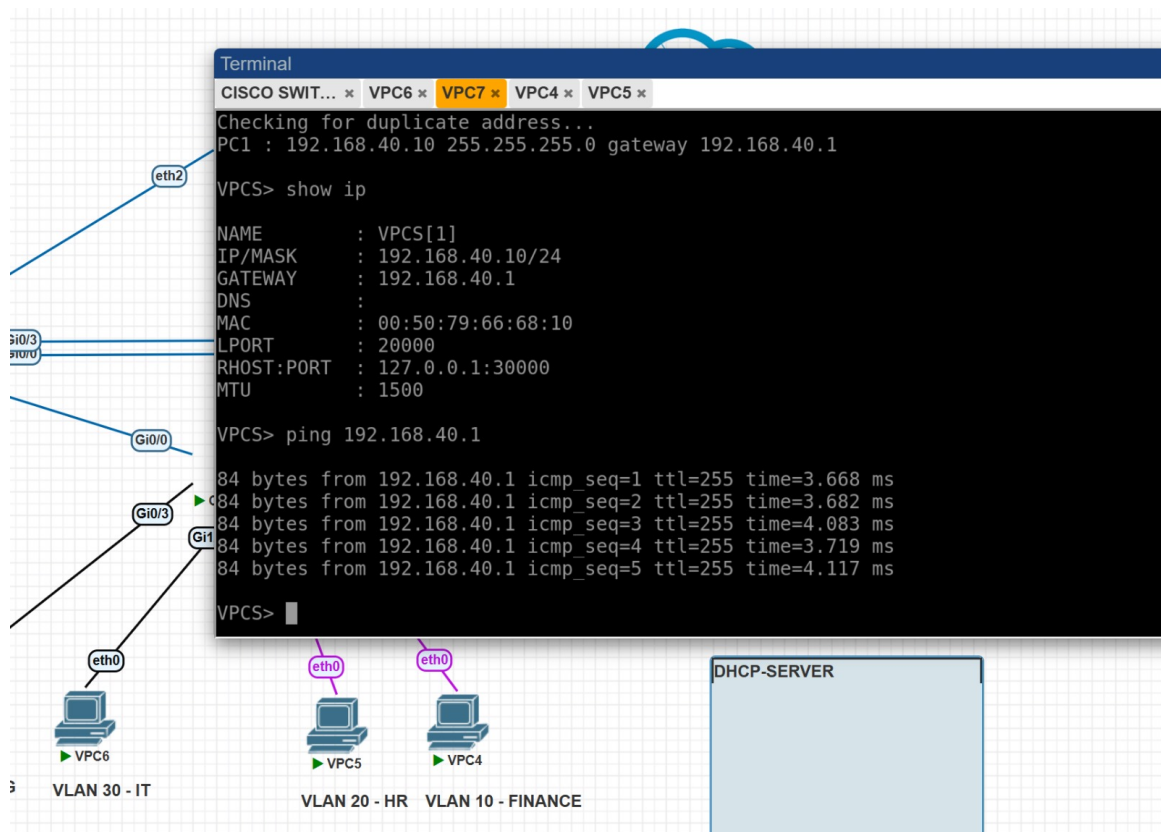
Pool VLAN10-FINANCE :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)        : 0 / 0
  Total addresses                  : 254
  Leased addresses                 : 1
  Pending event                   : none
  1 subnet is currently in the pool :
  Current index    IP address range    Leased addresses
  192.168.10.12    192.168.10.1 - 192.168.10.254    1

Pool VLAN20-HR :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)        : 0 / 0
  Total addresses                  : 254
  Leased addresses                 : 1
  Pending event                   : none
  1 subnet is currently in the pool :
  Current index    IP address range    Leased addresses
  192.168.20.12    192.168.20.1 - 192.168.20.254    1
Router#
Router#
```

```
Current index    IP address range    Leased addresses
192.168.20.12    192.168.20.1 - 192.168.20.254    1
Router#
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address        Client-ID/          Lease expiration    Type
                  Hardware address/
                  User name
192.168.10.11     0100.5079.6668.0d   Jun 07 2026 03:42 AM Automatic
192.168.20.11     0100.5079.6668.0e   Jun 07 2026 03:42 AM Automatic
Router#
Router#
```







```

VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=9.996 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=8.235 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=6.720 ms
^C
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=10.568 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=5.978 ms
^C
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=12.027 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=4.487 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=6.171 ms
^C
VPCS>

```

```

Router#
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.0d  Jun 07 2026 03:42 AM    Automatic
192.168.20.11    0100.5079.6668.0e  Jun 07 2026 03:42 AM    Automatic
Router#
Router#

```

```

Router#
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.0d  Jun 07 2026 03:42 AM    Automatic
192.168.20.11    0100.5079.6668.0e  Jun 07 2026 03:42 AM    Automatic
Router#
Router#

```

```

Press '?' to get help.
OS Cl
VPCS>
VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 192.168.50.10/24
GATEWAY     : 192.168.50.1
DNS         :
MAC         : 00:50:79:66:68:12
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500

VPCS>

```

```

VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=10.434 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=5.378 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=2.991 ms
^C
VPCS>

```

```

Router#
Router#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.10.11.1	YES	manual	up	up
GigabitEthernet0/1	10.10.30.1	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	up	up
GigabitEthernet0/2.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/2.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/2.30	192.168.30.1	YES	manual	up	up
GigabitEthernet0/2.40	192.168.40.1	YES	manual	up	up
GigabitEthernet0/3	10.10.10.1	YES	manual	up	up

```

Router#
Router#

```

```

new password:
repeat new password>
[admin@MikroTik] > /ip address add address=10.10.20.2/30 interface=ether1 comment=TO-HUAWEI
[admin@MikroTik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;; TO-HUAWEI
10.10.20.2/30 10.10.20.0 ether1
1 ;; TO-CISCO
10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] >

```

```

GigabitEthernet0/3 10.10.10.1 YES manual up
Router#
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/7 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/2/3 ms
Router#

```

```

<HUAWEI>
<HUAWEI>ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.10.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms

<HUAWEI>
<HUAWEI>ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.11.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms

<HUAWEI>
<HUAWEI>ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
  Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=2 ms
  Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=2 ms
  Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

--- 10.10.20.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/2 ms

<HUAWEI>
<HUAWEI>

```

```
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.20.1                            56 255 4ms
1 10.10.20.1                            56 255 1ms
2 10.10.20.1                            56 255 1ms
3 10.10.20.1                            56 255 1ms
4 10.10.20.1                            56 255 1ms
5 10.10.20.1                            56 255 1ms
sent=6 received=6 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=4ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.30.1                            56 255 2ms
1 10.10.30.1                            56 255 1ms
sent=2 received=2 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms

[admin@MikroTik] > |
```

```
Gateway of last resort is not set

Router#
Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0

Router#
```

```

CISCO ROUTER x CISCO SWITCH x HUAWEI PC x HUAWEI ROUTER x MIKROTIK x + -
[*HUAWEI]commit
[~HUAWEI]
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
State: Full      Mode:Nbr is Slave      Priority: 1
DR: 10.10.10.1      BDR: 10.10.10.2      MTU: 1500
Dead timer due in 38 sec
Retrans timer interval: 5
Neighbor is up for 00h00m05s
Neighbor Up Time : 2026-06-06 04:12:04
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
State: Full      Mode:Nbr is Slave      Priority: 1
DR: 10.10.11.1      BDR: None      MTU: 1500
Dead timer due in 36 sec
Retrans timer interval: 5
Neighbor is up for 00h00m05s
Neighbor Up Time : 2026-06-06 04:12:04
Authentication Sequence: [ 0 ]

[~HUAWEI]
[~HUAWEI]
[~HUAWEI]

```

```

Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
2.2.2.2          1    FULL/BDR        00:00:36    10.10.11.2   GigabitEthernet0/0
2.2.2.2          1    FULL/BDR        00:00:38    10.10.10.2   GigabitEthernet0/3
Router#
Router#

```

```

Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

 10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O       10.10.20.0/30 [110/110] via 10.10.11.2, 00:00:38, GigabitEthernet0/0
          [110/110] via 10.10.10.2, 00:00:38, GigabitEthernet0/3
O       192.168.50.0/24 [110/11] via 10.10.11.2, 00:00:38, GigabitEthernet0/0
          [110/11] via 10.10.10.2, 00:00:38, GigabitEthernet0/3
Router#
Router#

```



```

CISCO ROUTER x CISCO SWITCH x HUAWEI PC x HUAWEI ROUTER x MIKROTIK x + v - □ x
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
_public_ Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask    Proto  Pre  Cost    Flags NextHop         Interface
10.10.30.0/30       OSPF   10   110      D  10.10.11.1 Ethernet1/0/4
                   OSPF   10   110      D  10.10.10.1 Ethernet1/0/0
192.168.10.0/24      OSPF   10   11        D  10.10.11.1 Ethernet1/0/4
                   OSPF   10   11        D  10.10.10.1 Ethernet1/0/0
192.168.20.0/24      OSPF   10   11        D  10.10.11.1 Ethernet1/0/4
                   OSPF   10   11        D  10.10.10.1 Ethernet1/0/0
192.168.30.0/24      OSPF   10   11        D  10.10.11.1 Ethernet1/0/4
                   OSPF   10   11        D  10.10.10.1 Ethernet1/0/0
192.168.40.0/24      OSPF   10   11        D  10.10.11.1 Ethernet1/0/4
                   OSPF   10   11        D  10.10.10.1 Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask    Proto  Pre  Cost    Flags NextHop         Interface
10.10.10.0/30       OSPF   10   10        10.10.10.2 Ethernet1/0/0
10.10.11.0/30       OSPF   10   10        10.10.11.2 Ethernet1/0/4
10.10.20.0/30       OSPF   10  100       10.10.20.1 Ethernet1/0/3
192.168.50.0/24     OSPF   10   1         192.168.50.1 Ethernet1/0/2

[~HUAWEI]
[~HUAWEI]

```

```

[~HUAWEI]display ospf routing

      OSPF Process 1 with Router ID 2.2.2.2
      Routing Tables

Routing for Network
Destination      Cost    Type    NextHop    AdvRouter    Area
10.10.10.0/30    10      Direct  10.10.10.2  2.2.2.2      0.0.0.0
10.10.11.0/30    10      Direct  10.10.11.2  2.2.2.2      0.0.0.0
10.10.20.0/30    100     Direct  10.10.20.1  2.2.2.2      0.0.0.0
10.10.30.0/30    110     Stub    10.10.11.1  1.1.1.1      0.0.0.0
10.10.30.0/30    110     Stub    10.10.10.1  1.1.1.1      0.0.0.0
192.168.10.0/24   11      Stub    10.10.11.1  1.1.1.1      0.0.0.0
192.168.10.0/24   11      Stub    10.10.10.1  1.1.1.1      0.0.0.0
192.168.20.0/24   11      Stub    10.10.11.1  1.1.1.1      0.0.0.0
192.168.20.0/24   11      Stub    10.10.10.1  1.1.1.1      0.0.0.0
192.168.30.0/24   11      Stub    10.10.11.1  1.1.1.1      0.0.0.0
192.168.30.0/24   11      Stub    10.10.10.1  1.1.1.1      0.0.0.0
192.168.40.0/24   11      Stub    10.10.11.1  1.1.1.1      0.0.0.0
192.168.40.0/24   11      Stub    10.10.10.1  1.1.1.1      0.0.0.0
192.168.50.0/24   1       Direct  192.168.50.1 2.2.2.2      0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

      OSPF Process 65534 with Router ID 128.1.138.137
      Routing Tables

Routing for Network
Destination      Cost    Type    NextHop    AdvRouter    Area
128.1.138.137/32  0       Direct  128.1.138.137 128.1.138.137 0.0.0.0

Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0
[~HUAWEI]

```

```

[admin@MikroTik] > /routing ospf instance set default router-id=3.3.3.3
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=0.0.0.0 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2s
1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
  adjacency=10s
[admin@MikroTik] > |

```

```
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADo 10.10.10.0/30 10.10.20.1 110
10.10.30.1
1 ADo 10.10.11.0/30 10.10.20.1 110
10.10.30.1
2 ADo 192.168.10.0/24 10.10.30.1 110
3 ADo 192.168.20.0/24 10.10.30.1 110
4 ADo 192.168.30.0/24 10.10.30.1 110
5 ADo 192.168.40.0/24 10.10.30.1 110
6 ADo 192.168.50.0/24 10.10.20.1 110
[admin@MikroTik] > |
```

```
Router#
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:37 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/BDR 00:00:39 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/BDR 00:00:33 10.10.10.2 GigabitEthernet0/3
Router#
Router#|
```

```
CISCO ROUTER X CISCO SWITCH X HUAWEI PC X HUAWEI ROUTER X MIKROTIK X + v

OSPF Process 1 with Router ID 2.2.2.2
Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 35 sec
Retrans timer interval: 5
Neighbor is up for 00h05m00s
Neighbor Up Time : 2026-06-06 04:12:04
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00h00m53s
Neighbor Up Time : 2026-06-06 04:16:11
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 33 sec
Retrans timer interval: 5
Neighbor is up for 00h05m00s
Neighbor Up Time : 2026-06-06 04:12:04
Authentication Sequence: [ 0 ]

<HUAWEI>|
```

```
6 ADo 192.168.50.0/24 10.10.20.1 110
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
backup-dr-address=10.10.20.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
adjacency=1m20s
1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
adjacency=1m28s
[admin@MikroTik] > |
```

```
Router#
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:05:50 ago
Routing Descriptor Blocks:
10.10.11.2, from 2.2.2.2, 00:05:50 ago, via GigabitEthernet0/0
Route metric is 11, traffic share count is 1
* 10.10.10.2, from 2.2.2.2, 00:05:50 ago, via GigabitEthernet0/3
Route metric is 11, traffic share count is 1
Router#
Router#|
```



```

CISCO ROUTER  x  CISCO SWITCH  x  HUAWEI PCJ  x  HUAWEI ROUTER  x  MIKROTIK  x  +
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:05:50 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:05:50 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:05:50 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O       10.10.20.0/30 [110/110] via 10.10.11.2, 00:06:05, GigabitEthernet0/0
        [110/110] via 10.10.10.2, 00:06:05, GigabitEthernet0/3
O       192.168.50.0/24 [110/11] via 10.10.11.2, 00:06:05, GigabitEthernet0/0
        [110/11] via 10.10.10.2, 00:06:05, GigabitEthernet0/3
Router#
Router#

```

```

CISCO ROUTER  x  CISCO SWITCH  x  HUAWEI PCJ  x  HUAWEI ROUTER  x  +
Error: Unrecognized command found at '^' position.
<HUAWEI>
<HUAWEI>display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
_public_ Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask    Proto    Pre  Cost    Flags NextHop          Interface
-----
10.10.30.0/30      OSPF     10   110      D  10.10.11.1  Ethernet1/0/4
10.10.30.0/30      OSPF     10   110      D  10.10.10.1  Ethernet1/0/0
192.168.10.0/24     OSPF     10   11        D  10.10.11.1  Ethernet1/0/4
192.168.10.0/24     OSPF     10   11        D  10.10.10.1  Ethernet1/0/0
192.168.20.0/24     OSPF     10   11        D  10.10.11.1  Ethernet1/0/4
192.168.20.0/24     OSPF     10   11        D  10.10.10.1  Ethernet1/0/0
192.168.30.0/24     OSPF     10   11        D  10.10.11.1  Ethernet1/0/4
192.168.30.0/24     OSPF     10   11        D  10.10.10.1  Ethernet1/0/0
192.168.40.0/24     OSPF     10   11        D  10.10.11.1  Ethernet1/0/4
192.168.40.0/24     OSPF     10   11        D  10.10.10.1  Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask    Proto    Pre  Cost    Flags NextHop          Interface
-----
10.10.10.0/30       OSPF     10   10        10.10.10.2  Ethernet1/0/0
10.10.11.0/30       OSPF     10   10        10.10.11.2  Ethernet1/0/4
10.10.20.0/30       OSPF     10  100      10.10.20.1  Ethernet1/0/3
192.168.50.0/24     OSPF     10   1         192.168.50.1 Ethernet1/0/2
<HUAWEI>
<HUAWEI>
<HUAWEI>
<HUAWEI>

```

```

[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#   DST-ADDRESS      PREF-SRC  GATEWAY          DISTANCE
0  ADo  10.10.10.0/30      10.10.20.1      110
   ADo  10.10.11.0/30      10.10.30.1      110
1  ADo  10.10.11.0/30      10.10.20.1      110
   ADo  10.10.30.1         10.10.30.1      110
2  ADo  192.168.10.0/24     10.10.30.1      110
3  ADo  192.168.20.0/24     10.10.30.1      110
4  ADo  192.168.30.0/24     10.10.30.1      110
5  ADo  192.168.40.0/24     10.10.30.1      110
6  ADo  192.168.50.0/24     10.10.20.1      110
[admin@MikroTik] >

```

```
VLAN10
0;VPC4
VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=254 time=7.133 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=254 time=5.788 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=254 time=4.602 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=254 time=5.584 ms
^C
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=10.519 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=8.501 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=8.377 ms
^C
VPCS> |
```

```
VLAN20
0;VPC5
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.065 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=6.257 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.831 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=8.408 ms
^C
VPCS> |
```

```
VLAN30
0;VPC6
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.550 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.185 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.030 ms
^C
VPCS> |
```

```
VLAN40
0;VPC7
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.291 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=6.187 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.197 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=10.577 ms
^C
VPCS> |
```

```
CISCO ROUTER  X CISCO SWITCH  X HUAWEI PCJ  X HUAWEI ROUTER  X
                                OSPF    10    11                D    10.10.10.1    Ethernet1/
OSPF routing table status : <Inactive>
    Destinations : 4          Routes : 4

Destination/Mask    Proto    Pre    Cost           Flags NextHop           Interface
-----
10.10.10.0/30       OSPF     10     10              10.10.10.2       Ethernet1/
10.10.11.0/30       OSPF     10     10              10.10.11.2       Ethernet1/
10.10.20.0/30       OSPF     10     100             10.10.20.1       Ethernet1/
192.168.50.0/24     OSPF     10     1               192.168.50.1     Ethernet1/

<HUAWEI>
<HUAWEI>
<HUAWEI>
<HUAWEI>ping 192.168.10.11
  PING 192.168.10.11: 56 data bytes, press CTRL_C to break
    Reply from 192.168.10.11: bytes=56 Sequence=1 ttl=63 time=4 ms
    Reply from 192.168.10.11: bytes=56 Sequence=2 ttl=63 time=4 ms
    Reply from 192.168.10.11: bytes=56 Sequence=3 ttl=63 time=4 ms
    Reply from 192.168.10.11: bytes=56 Sequence=4 ttl=63 time=5 ms
    Reply from 192.168.10.11: bytes=56 Sequence=5 ttl=63 time=4 ms

  --- 192.168.10.11 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 4/4/5 ms

<HUAWEI>
<HUAWEI>|
```

```
<HUAWEI>ping 192.168.20.11
PING 192.168.20.11: 56 data bytes, press CTRL_C to break
  Reply from 192.168.20.11: bytes=56 Sequence=1 ttl=63 time=6 ms
  Reply from 192.168.20.11: bytes=56 Sequence=2 ttl=63 time=4 ms
  Reply from 192.168.20.11: bytes=56 Sequence=3 ttl=63 time=6 ms
  Reply from 192.168.20.11: bytes=56 Sequence=4 ttl=63 time=18 ms
  Reply from 192.168.20.11: bytes=56 Sequence=5 ttl=63 time=4 ms

--- 192.168.20.11 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 4/7/18 ms

<HUAWEI>
<HUAWEI>
```

```
<HUAWEI>ping 192.168.30.10
PING 192.168.30.10: 56 data bytes, press CTRL_C to break
  Reply from 192.168.30.10: bytes=56 Sequence=1 ttl=63 time=6 ms
  Reply from 192.168.30.10: bytes=56 Sequence=2 ttl=63 time=4 ms
  Reply from 192.168.30.10: bytes=56 Sequence=3 ttl=63 time=4 ms
  Reply from 192.168.30.10: bytes=56 Sequence=4 ttl=63 time=4 ms
  Reply from 192.168.30.10: bytes=56 Sequence=5 ttl=63 time=6 ms

--- 192.168.30.10 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 4/4/6 ms

<HUAWEI>
<HUAWEI>
```

```
<HUAWEI>ping 192.168.40.10
PING 192.168.40.10: 56 data bytes, press CTRL_C to break
  Reply from 192.168.40.10: bytes=56 Sequence=1 ttl=63 time=4 ms
  Reply from 192.168.40.10: bytes=56 Sequence=2 ttl=63 time=7 ms
  Reply from 192.168.40.10: bytes=56 Sequence=3 ttl=63 time=4 ms
  Reply from 192.168.40.10: bytes=56 Sequence=4 ttl=63 time=6 ms
  Reply from 192.168.40.10: bytes=56 Sequence=5 ttl=63 time=6 ms

--- 192.168.40.10 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 4/5/7 ms

<HUAWEI>
<HUAWEI>
```

```

VLAN10
0;VPC4
VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=254 time=7.133 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=254 time=5.788 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=254 time=4.602 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=254 time=5.584 ms
^C
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=10.519 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=8.501 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=8.377 ms
^C
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    6.215 ms  3.868 ms  3.852 ms
 2  10.10.11.2     5.338 ms  5.235 ms  5.901 ms
 3  *192.168.50.10 8.304 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> |

```

```

Router(config)#do show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:14:14 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:14:14 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
Router(config)#
Router(config)#

```

```

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    5.552 ms  3.380 ms  4.727 ms
 2  10.10.11.2     7.359 ms  5.610 ms  4.944 ms
 3  *192.168.50.10 10.154 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> |

```

```

Router(config)#do show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:00:13 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:00:13 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
Router(config)#
Router(config)#

```

```

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    3.614 ms  3.631 ms  4.510 ms
 2  10.10.30.2     4.652 ms  3.447 ms  3.378 ms
 3  10.10.20.1    11.464 ms  6.592 ms  5.997 ms
 4  *192.168.50.10 6.087 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> |

```



```

Router(config)#do show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:00:38 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:00:38 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:00:38 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router(config)#
Router(config)#

```

```

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    4.942 ms  5.270 ms  4.586 ms
 2  10.10.11.2     7.938 ms  5.845 ms  5.319 ms
 3  *192.168.50.10 10.986 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>

```

```

Switch#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2, Gi0/3 Gi1/0, Gi1/1, Gi1/2, Gi1/3
10	FINANCE	active	
20	HR	active	
30	IT	active	
40	MARKETING	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```

Switch#
Switch#

```

```

*Jun  6 03:33:49.589: %SYS-5-CONFIG_I: Configured from console

```

```

Switch#

```

```

Switch#show interfaces trunk

```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1
Port	Vlans allowed on trunk			
Gi0/0	10,20,30,40			
Port	Vlans allowed and active in management domain			
Gi0/0	10,20,30,40			
Port	Vlans in spanning tree forwarding state and not pruned			
Gi0/0	10,20,30,40			

```

Switch#
Switch#

```